OKRA TECH LABS

AI & Enterprise Software Engineering

PROPOSAL & QUOTATION

Project Nexus — Enterprise RAG AI Assistant

Proof-of-Concept Engagement

Prepared for	**Plutomen**	**Prepared by**	Okra Tech Labs
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1. Engagement Overview

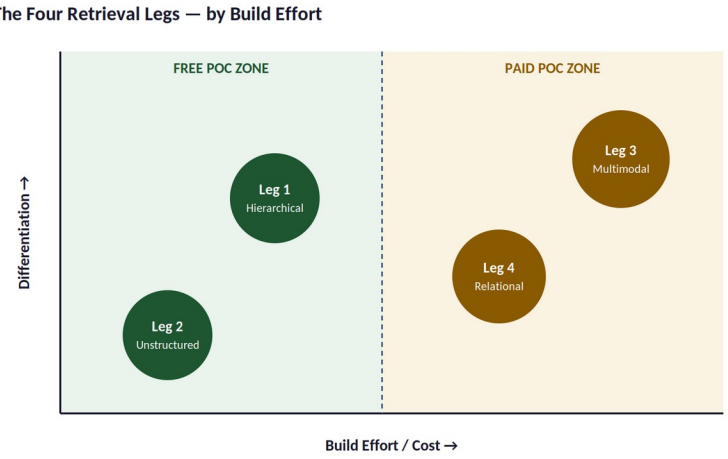
Plutomen has shared the Product Requirements Document for **Project Nexus**, a multimodal Retrieval-Augmented Generation (RAG) assistant intended to serve as a unified, grounded intelligence layer over proprietary enterprise data — Standard Operating Procedures, Inspection Reports, and a multi-format Document Library.

On close reading, the PRD does not describe a single RAG system. It bundles **four distinct retrieval strategies** under one interface, each with different engineering characteristics. To de-risk the build and give Plutomen a working system to evaluate quickly, Okra Tech Labs proposes a phased proof-of-concept structured around these strategies.

The four retrieval strategies within Project Nexus

#	Retrieval Strategy	What it does
1	**Structured / Hierarchical RAG**	Retrieval that respects the SOP Section→Step tree and checklist Page→Section→Question structure — returns a step in its proper context, not a floating chunk.
2	**Classic Unstructured RAG**	Hybrid keyword + semantic search over the free-form Document Library (PDF, XLS, CSV, TXT). The conventional RAG most people mean by the term.
3	**Multimodal RAG**	Cross-modal retrieval over images and video — photo-in to SOP-step-out, and keyframe/transcript search over MP4 evidence.
4	**Relational / Metadata RAG**	Structured queries over Findings/Actions objects (assignee, site, date, status) — e.g. “last 5 reports by Frank.” Closer to text-to-query than to embeddings; needs a query-routing layer.

Legs 1 and 2 share a common ingestion-and-retrieval foundation and can be demonstrated quickly. Legs 3 and 4 each require dedicated pipelines — vision/video processing for the former, a query router and structured-filter engine for the latter — and carry materially higher build cost. The phasing below reflects that split.



Positioning of the four legs — the two lower-effort, foundational legs form the complimentary POC; the two higher-effort, differentiating legs form the paid POC.

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2. Phase A — Complimentary Proof of Concept

Cost: ₹0 — Complimentary	**Timeline: 2 weeks** from commencement of work

Okra Tech Labs will build and demonstrate a working RAG assistant covering the two foundational retrieval legs, grounded on a representative sample of Plutomen's SOP and document data.

In scope

- **Structured / Hierarchical RAG (Leg 1):** ingestion pipeline preserving the SOP Section→Step tree; retrieval that returns steps in hierarchical context.
- **Classic Unstructured RAG (Leg 2):** ingestion of text-based document formats (PDF, TXT, and text extracted from XLS/CSV) with hybrid BM25 + dense-vector search.
- **Grounded generation with citations:** answers anchored to source chunks with explicit references (e.g. [SOP-102, Section 4, Step 2.1]).
- **Graceful fallback message** when no confident match is found, per PRD §4.3.
- **Multi-turn conversational context** for follow-up questions within a session.
- **A functional demo interface** to run live queries against the sample corpus.

Explicitly out of scope for Phase A

- Multimodal ingestion — image, and video (Leg 3).
- Relational / metadata querying over Findings/Actions (Leg 4).
- Production hardening: multi-tenant isolation, folder/role ACL security trimming, token-metering ledger, rate limiting, and the 500-concurrent-session scalability target.
- Feedback-capture module and retraining store (§4.4).

These are addressed in Phase B and in the subsequent production engagement.

What Okra Tech Labs needs from Plutomen to begin

- A representative sample set of SOPs and library documents (anonymised where required).
- A short list of 8–10 real questions the assistant should be able to answer, for acceptance.
- A single point of contact for clarifications during the 2-week window.

The 2-week clock starts on the day active development begins — after sample data is received — and does not count scoping calls or meetings.

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3. Phase B — Paid Proof of Concept

Cost: ₹75,000 (partner rate)	**Timeline:** 5–6 weeks

Phase B extends the working POC to the two advanced retrieval strategies — the legs that carry the real engineering weight and that most clearly differentiate Project Nexus from a generic assistant.

In scope

- **Multimodal RAG (Leg 3):** vision-based ingestion of images embedded in SOP steps and uploaded as inspection evidence (via a vision-language model), plus video processing through keyframe sampling and/or audio-transcript extraction, making visual and video content searchable in natural language.
- **Relational / Metadata RAG (Leg 4):** a query-routing layer that recognises structured questions and resolves them against Findings/Actions objects — assignee, site, date, and status filters — enabling queries such as "show all failed safety checks for Zone B last quarter" or "last 5 reports submitted by Frank."
- **Cross-modal synthesis** combining the above with the Phase A legs under one query interface.

Why Phase B is a paid engagement

The multimodal pipeline introduces vision-language and video-processing compute that the two foundational legs do not require, and the relational leg requires an entirely separate query-routing and structured-filter engine rather than embedding-based search. Each represents dedicated build effort beyond the complimentary scope. The rate above is offered as a partner rate in view of a longer-term working relationship.

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4. Commercial Summary

Phase	Scope	Timeline	Cost
Phase A — Complimentary POC	Legs 1 & 2 — Structured + Unstructured RAG	2 weeks	**₹0**
Phase B — Paid POC	Legs 3 & 4 — Multimodal + Relational RAG	5–6 weeks	**₹75,000**
Production build	Full platform — security, governance, scale (§5, §6)	*Scoped separately*	TBD

The full production system described in the PRD — multi-tenant isolation, ACL security trimming, the token-usage ledger and rate-limiting engine, the feedback/retraining store, and the 500-concurrent-session performance target — sits beyond both POC phases and will be scoped and quoted separately once the POC has validated the approach.





Timeline is sequential — Phase B begins once Phase A is complete. Investment shown at partner rate; production build scoped separately.

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5. Technical Approach

This section outlines *how* Okra Tech Labs intends to build Project Nexus, at a level sufficient for technical review. Specific frameworks, libraries, and model providers will be finalised during the engagement based on Plutomen's environment, data characteristics, and cost constraints; the process below is provider-agnostic.

5.1 Ingestion & indexing pipeline

Source content is processed through a structured pipeline before it can be queried:

- **Parsing & structure preservation:** each source type is parsed with a strategy that preserves its inherent structure — the SOP Section→Step hierarchy and checklist templates are ingested as trees rather than flattened text, so retrieval can later return a step in its correct context.
- **Chunking:** documents are segmented into semantically coherent units sized for retrieval, with overlap tuned to avoid splitting related instructions across chunk boundaries.
- **Metadata tagging:** every chunk is tagged at ingestion with source identifiers, hierarchy position, and access-control attributes, so that downstream filtering (security and, later, multi-tenant isolation) can be enforced at query time.
- **Embedding & vector index:** chunks are converted into vector representations using an embedding model and stored in a vector index, alongside a keyword index — enabling the hybrid retrieval described below.

5.2 Retrieval & grounded generation

- **Hybrid retrieval:** queries are served by combining exact keyword matching (for precise terms such as asset serial numbers) with semantic vector search (for meaning-based matches), then merging and re-ranking the results — capturing both literal and conceptual relevance.
- **Context assembly:** the highest-ranked chunks, with their source metadata, are assembled into a bounded context window and passed to the generation model.
- **Grounded answer generation:** the model is instructed to answer strictly from the retrieved context and to attach explicit source citations. Where the retrieved context is insufficient or conflicting, the system returns a graceful fallback and proactive suggestions rather than an unverified answer.

- **Verification step:** generated answers are checked against the retrieved source chunks before being returned, to catch and suppress any statement not supported by the source material — the core anti-hallucination safeguard.

5.3 Multimodal processing (Phase B)

Visual and video content is made searchable by converting it into retrievable representations during ingestion: images are described and/or embedded via a vision-capable model; videos are broken into sampled keyframes and, where audio is present, transcribed — so that a natural-language query can surface the relevant image or moment in a video. This stage is the primary driver of the compute requirement noted in Section 6.

5.4 Relational querying (Phase B)

Questions that are structured rather than semantic — for example, filtering inspection findings by assignee, site, date, or status — are handled by a routing layer that recognises the intent and resolves it against structured records, instead of forcing every question through semantic search. This is what allows precise queries such as retrieving the last several reports submitted by a specific user.

5.5 Targets requiring validation

Several success metrics and non-functional targets in the PRD are ambitious and are best treated as **aspirational goals to be validated and calibrated during the POC**, rather than guaranteed acceptance thresholds. Confirming what is achievable — and at what infrastructure cost — is itself a core value of the proof-of-concept. Specifically:

- **≥99% grounding / hallucination reduction (§2.2):** near-perfect grounding is only meaningful against an agreed, labelled evaluation set. We recommend jointly defining that evaluation set early; the achievable figure will be measured, not assumed, and "zero hallucination" cannot be guaranteed as an absolute.
- **P95 latency ≤3.0s (text) / ≤6.0s (multimodal) (§2.2):** these targets are in tension with the dual-stage verification and anti-hallucination checks also required in §4.3 — each verification pass adds latency. Meeting both simultaneously depends heavily on model choice and hardware; a realistic latency-versus-rigour trade-off will be established during the POC.
- **500 concurrent sessions without degradation (§5.3):** sustained concurrency at this level is a function of provisioned compute (GPU/inference capacity), not application logic alone. It is a production-scale target that sits beyond the POC and will require appropriately sized infrastructure to guarantee.
- **New content queryable within 5 minutes of upload (§5.3):** end-to-end freshness (parsing, embedding, indexing — and for media, vision/video processing) within five minutes is achievable at modest volumes but becomes infrastructure-dependent as corpus size and upload frequency grow.
- **≥85% fallback relevance acceptance (§2.2):** a reasonable target, but one that can only be confirmed against real user queries and real data during evaluation.

None of the above changes the proposed scope — they are noted so that expectations are set transparently and the POC can produce honest, measured figures against each target.

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6. Assumptions & Notes

- Both phases are proof-of-concept engagements intended to validate approach and quality on representative data — not production deployments.
- Timelines assume timely provision of sample data, access, and clarifications by Plutomen.
- **Hosting environment (Phase A):** Okra Tech Labs can host the complimentary POC (Legs 1 & 2) on its own servers as an interim trial for evaluation, with the solution subsequently handed over and integrated into Plutomen's application (see delivery note below).
- **Hosting environment (Phase B):** the paid POC introduces multimodal processing (vision and video), which is compute-intensive by nature. For this phase the underlying cloud infrastructure — including

GPU/compute for vision and video workloads (e.g. AWS or equivalent) — is to be provisioned on Plutomen's own environment, so that the proof-of-concept runs on the same footing as the intended production setup. Okra Tech Labs will build, configure, and deploy into that environment.

- **Delivery & integration:** each phase can be handed over and packaged for binding into Plutomen's existing application. Application-side integration effort will be confirmed against Plutomen's app architecture once shared.
- Third-party costs (LLM/API usage, cloud infrastructure) during the POC are minimal at sample scale; any material usage will be discussed in advance.
- This quotation is valid for 30 days from the date of issue.